Description

You can use this inbound service provided by SAP Business Network for Logistics to cancel a freight order for subcontracting from your external system, such as a transportation management system. This request will be presented to your carriers.

Structure

For more information, see: Request Transportation Order Cancellation

Consumer API to Confirm Freight Order for Subcontracting

As a shipper, you can confirm a freight order for subcontracting in your transportation management system.

Technical Name

`TransportationOrderConfirmation_In`

Description

You can use this outbound service consumed by SAP Business Network for Logistics to confirm a freight order for subcontracting in your external system, such as a transportation management system. The confirmation will be sent by your carrier.

Additional Information

- Sending remarks to the carrier

  You can use the standard text type DFLT to send remarks to carrier.

- Maintaining package descriptions

  In the **Assign Codes** app, you can specify the package material type descriptions. On the left pane, in the **Items** section, choose the **Packaging Material** Type. On the right pane, specify the packaging codes in your external transportation system and the corresponding **LBN Code**.

Maintaining Standard Carrier Alpha Code (SCAC)

The SCAC is a standardized code to identify carriers in transportation processes. In SAP Transportation Management it is used in freight orders and freight bookings to identify respective carriers.

You can assign SCAC code to a carrier using transaction BP. Select the business partner role as carrier. In the **Identification** tab, under **Identification Numbers** section, you can specify the SCAC code. In the **IDType** column, from the value help, choose BUP006, which is the Standard Carrier Alpha Code. Specify an identification number.

Once you assign a SCAC to a carrier, you can now find your carrier in your transportation documents such as the freight order, freight booking, and so on.

Structure

For more information, see: Change Transportation Order Based on Transportation Order Confirmation

Freight Booking

Using the freight booking APIs, you as a shipper can confirm, create, notify, or cancel freight bookings when integrating SAP Business Network for Logistics with external systems.

You can perform the following activities using the APIs for freight booking:

- You can use the Provider API to Create Freight Booking to create a freight booking request in SAP Business Network for Logistics.
- You can use the Provider API to Cancel Freight Booking to cancel a freight booking request in SAP Business Network for Logistics.
- You can use the Provider API for Freight Booking Waybill Notification to send a notification about a waybill that is to be issued by a carrier. The information in the notification enables a carrier to issue a waybill with reference to a confirmation of a freight booking.
- You can use the Consumer API to Confirm Freight Booking to confirm a freight booking request in your transportation management system.

Provider API to Create Freight Booking

As a shipper, you can create a freight booking request in SAP Business Network for Logistics.

Technical Name

`TransportationOrderBookingRequest_Out`

Description

You can use this inbound service to create a freight booking request in your SAP Business Network for Logistics based on the information from your external system, such as a transportation management system. This request will be presented to your carriers.

Note that SAP Business Network for Logistics accepts only 700 items in a freight booking document.

Structure

For more information, see, Request Transportation Order Booking.

Provider API to Cancel Freight Booking

As a shipper, you can cancel a freight booking request in SAP Business Network for Logistics.

Technical Name

`TransportationOrderBookingCancellationRequest_Out`

Description

You can use this inbound service to cancel a freight booking in your SAP Business Network for Logistics based on the information from your external system, such as a transportation management system. This request will be presented to your carriers.

Structure

For more information, see, Request Transportation Order Booking Cancellation.

Provider API for Freight Booking Waybill Notification

As a shipper, you can send a notification about a waybill that is to be issued by a carrier. The information in the notification enables a carrier to issue a waybill with reference to a confirmation of a freight booking.

Technical Name

`TransportationOrderBookingWaybillNotification_Out`

Description

You can use this inbound service to enable a freight booking waybill notification request in your SAP Logistics Business Network based on the information from your external system, such as a transportation management system. This request will be presented to your carriers.

Structure

For more information, see, https://help.sap.com/viewer/14e2fd56dd2e43b1b9db5120dcd89bae/9.6.2/en-US/2e9019f2681211df375c000f20fcb6a9.html.

Consumer API to Confirm Freight Booking

As a shipper, you can confirm a freight booking request in your transportation management system.

Technical Name

`TransportationOrderBookingConfirmation_In`

Description

You can use this outbound service to confirm a freight booking in your external system such as a transportation management system, based on the information provided by SAP Business Network for Logistics. The confirmation will be sent by your carrier.

Additional Information

- Sending remarks to the carrier

  You can use the standard text type `DFLT` to send remarks to carrier.

- Maintaining package descriptions

In the **Assign Codes** app, you can specify the package material type descriptions. On the left pane, in the **Items** section, choose the **Packaging Material** Type. On the right pane, specify the packaging codes in your external transportation system and the corresponding **LBN Code**.

Maintaining Standard Carrier Alpha Code (SCAC)

The SCAC is a standardized code to identify carriers in transportation processes. In SAP Transportation Management it is used in freight orders and freight bookings to identify respective carriers.

You can assign SCAC code to a carrier using transaction BP. Select the business partner role as carrier. In the **Identification** tab, under **Identification Numbers** section, you can specify the SCAC code. In the **IDType** column, from the value help, choose `BUP006`, which is the Standard Carrier Alpha Code. Specify an identification number.

Once you assign a SCAC to a carrier, you can now find your carrier in your transportation documents such as the freight order, freight booking, and so on.

Structure

For more information, see, Change Order Based on Transportation Order Booking Confirmation.

Consumer API to Send Attachments

As a shipper, you can receive attachments in your Transportation Management on SAP S/4 HANA system from your SAP Business Network Freight Collaboration system.

Technical Name

`BNAttachmentForConfirmation_In`

Description

You can use this outbound service to receive attachments in your Transportation Management on SAP S/4 HANA system. You can use this interface to send attachments during the freight booking confirmation process. Note that sending of attachments is not supported in the existing SOAP API, Consumer API to Confirm Freight Booking.

Dock Appointment Scheduling

As a shipper, you can use dock appointment scheduling APIs in SAP Business Network Freight Collaboration to book dock appointments for a loading point on behalf of the carrier for loading or unloading services in the docking locations of your organization.

Using the dock appointment scheduling APIs, you can create, update, or cancel various dock appointment scheduling and gate processes.

Note

To avoid messages out of sequence or late runners overwriting newer updates, the processing of these late runner messages will be skipped.

The `MessageHeader → CreationDateTime` field will be used to determine if the update will be processed or skipped. In case the `CreationDateTime` will not be present, the message will always be processed as the latest update.

You can perform the following activities using the APIs for dock appointment scheduling:

- You can use the Provider API for Location Replication to replicate location master data in SAP Business Network for Logistics.
- You can use the Provider API for Dock Appointment Scheduling to create dock appointment requests and confirmed dock appointments in your SAP Business Network for Logistics.
- You can use the Provider API for Dock Appointment Cancellation to cancel the dock appointments in your SAP Business Network for Logistics.
- You can use the Consumer API for Dock Appointment Scheduling to update your external systems with the appointment details.
- You can use the Provider API to Manage Gate Operations to update the gate logbook details in your SAP Business Network for Logistics.
- You can use the Consumer API to Manage Gate Operations to update your external systems with the check-in and checkout details provided by SAP Business Network for Logistics.

More Information

Dock Appointment Scheduling

Provider API for Location Replication

As a shipper, you can replicate location master data in SAP Business Network for Logistics.

Technical Name

`LocationBulkReplicationRequest_Out`

Description

This asynchronous inbound service enables you to replicate location from an external system to SAP Business Network for Logistics.

Structure

The SOAP interface contains the business data for location replication.

Interface Node	Description
LocationBulkReplicateRequestMessage	Bulk Service
MessageHeader	Message Header for the Bulk Service
LocationReplicationRequestMessage	Per location, contains message header, location information and address

Interface Node	Description
MessageHeader	Header Information
Location	Location IDs, Location Type Code, Description, Alternative Identifiers
Address	Address Fields

Provider API for Dock Appointment Scheduling

As a shipper, you can use this inbound service to create dock appointment requests and confirmed dock appointments in SAP Business Network for Logistics based on the information from your external systems such as a transportation system, warehouse management system, yard management system and so on. You can also use this service to provide a guest user access to your customers and carriers who are not on SAP Business Network for Logistics and enable them to maintain the appointment details.

ℹ Note

To avoid messages out of sequence or late runners overwriting newer updates, the processing of these late runner messages will be skipped.

The `MessageHeader` → `CreationDateTime` field will be used to determine if the update will be processed or skipped. In case the `CreationDateTime` will not be present, the message will always be processed as the latest update.

Technical Name

`GenericAppointment_In`

Description

In general, you can create or update appointment requests with this API. In addition to that, you can create or update booked appointments with this API. You can also update appointment requests to booked appointments.

- Appointment requests are not yet scheduled appointments. They are assigned to a docking location and optionally also to a loading point but not to a time slot. Thus, they do not consume capacity on a loading point. Appointment requests have the appointment status as `01`.
- Booked appointments are scheduled appointments, which means they have been assigned to a time slot on a certain loading point and consume its capacity. Booked appointments have the appointment status as `03`.

Appointment requests can be changed to booked appointments either via this API or via the shipper UI (**Manage Dock Appointments** app) or the carrier UI (**Self- Book Dock Appointments** app) by assigning a time slot on a certain loading point to them.

In more detail, you can perform the following in SAP Business Network for Logistics using this API:

- Create an appointment request

  Note that to create an appointment request, you must send the appointment status as `01` from your external system to the network. You might want to provide a new appointment external ID at this point. The corresponding appointment ID for the new appointment request is generated in the system.

- Book an appointment

Note that to book an appointment, you must send the appointment status as `03` (or `02`) and the appointment key from your external system to the network.

- Update an appointment request

  Note that to update an appointment request, you must send the appointment status as `01` and the appointment key from your external system to the network.

- Update a booked appointment

  Note that to update a booked appointment, you must send the appointment status as `03` and the appointment key from your external system to the network.

Note

The network accepts only 700 cargo items in an API call.

Prerequisites

You have performed the following:

- You have created a shipper and carrier business partner in your SAP Business Network for Logistics and your external system.
- In your external system, you have the LBN ID and you have mapped the LBN ID with the business partner.
- You have created the location master data and corresponding loading point and docking location in SAP Business Network for Logistics. You must use this location ID and loading point ID in the webservice SOA Manager interface.

Appointment Key

Appointments can be uniquely identified by a combination of the sender system ID and the appointment external ID that was used to create them, or by the appointment ID. The appointment ID is generated as a consecutive number during the creation of an appointment within the network, and it is shown in the UI.

Structure

They system has different fields categorized under different nodes. Each node contains various fields. Below are some of the important fields displayed by the system:

Technical Name	Field	Description	Node
OrderingPartyLbnId	Ordering Party LBN ID*	Contains the ordering party ID maintained in SAP Business Network for Logistics	Appointment
CarrierLbnId	Carrier LBN ID*	Contains the carrier ID maintained in SAP Business Network for Logistics	Appointment
SenderSystemId	Sender System ID*	Contains the ID of sender system (see also AppointmentExternalId)	Appointments
AppointmentId	Appointment ID	A consecutive number that serves as a key for the	Appointment

Technical Name	Field	Description	Node
		appointment. It is sent back to the caller via `GenericAppointment_Out` after the creation of an appointment in the network.	
AppointmentStatusCode	Appointment Status Code	Contains one of the following status codes: 01 – Appointment Request Created 02 – Pending Confirmation 03 – Confirmed	Appointments
AppointmentExternalId	Appointment External ID	An externally provided key for the appointment. It is provided during creation and then used during update operations. The key is unique in combination with the sender system ID.	Appointment
DockingLocationId	Docking Location ID*	Represents the location ID assigned to a docking loaction in the **Manage Entities** app. Example: QM7_910:LBN_WDF (<SourceSystem>: <ExternalID>) For more information about locations, see: Viewing a Location.	Appointment
LoadingPointId	Loading Point ID	Represents the loading point ID defined in the **Manage Entities** app (for example: Loading_Point_ID_1). This field is not required for appointment request creation but is required when creating a booked appointment. If the loading point ID is provided already for the appointment request, it cannot be changed anymore during appointment booking via the shipper or carrier UI. However, it still can be changed by the current API.	Appointment
LoadingActivityType	Loading Activity Type	Contains one of the following activity types: 01 – Loading 02 – Unloading 03 – Loading/Unloading	Appointments

Technical Name	Field	Description	Node
RequestedStartDateTime	Requested Start Date Time	If an appointment is booked via this API with status `02` (*Pending Confirmation*), this field is used for the determination of the booked timeslot. If an appointment request with status `01` (*Appointment Request Created*) is created via this API, the requested dates are used to determine the planned dates.	Appointment
RequestedEndDateTime	Requested End Date Time	If an appointment is booked via this API with status `02` (*Pending Confirmation*), this field is used for the determination of the booked timeslot. If an appointment request with status `01` (*Appointment Request Created*) is created via this API, the requested dates are used to determine the planned dates.	Appointment
AcceptableStartDateTime	Acceptable Start Date Time	Acceptable time period corresponds to a broader horizon and overwrites the **Enable Appointment Date Change** setting for carriers in the **Manage Entities** app. This enables carriers to change the date of a dock appointment in the given acceptable time range before the cut-off time for carrier is reached.	Appointment
AcceptableEndDateTime	Acceptable End Date Time	Acceptable time period corresponds to a broader horizon and overwrites the **Enable Appointment Date Change** setting for carriers in the **Manage Entities** app. This enables carriers to change the date of a dock appointment in the given acceptable time range before the cut-off time for carriers is reached.	Appointment
AppointmentStartDateTime	Appointment Start Date Time	Used for the determination of the booked time slot in combination with appointment status `03`.	Appointment
AppointmentEndDateTime	Appointment End Date Time	Used for the determination of the booked time slot in	Appointment

Technical Name	Field	Description	Node
		combination with appointment status 03.	
DocumentId	Document ID	Contains the document ID of the source business document from your external system (For example, freight order, transport unit, yard order).	Appointment Reference Document
CargoItemDescription	Cargo Item Description	Contains the cargo item description	Appointment Cargo Item
DangerousGoodsAreContained	Dangerous Goods Are Contained	Contains the information whether dangerous goods are contained	Appointment Cargo Item
CreatedAt	Created At	Contains the date and time at which the note was created	Appointment Note
DocumentType	Document Type	Contains one of the following types of reference documents: 001 – Purchase Order 58 – Inbound Delivery 73 – Outbound Delivery 111– Freight Request for Quotation 114 – Sales Order 612 – Shipment Order 614 – Freight Order 1122 – Transportation Order (Transportation Management) 1311 – Transportation Unit Appointment (Extended Warehouse Management) YO – Yard Order (Yard Logistics)	Appointment Reference Document

Note that the fields marked with * are mandatory fields.

ℹ Note

- We do not support the extensibility feature with this service as of now.
- We do not support the consolidation of multiple document references to one appointment document.

Confirmed

If you book an appointment with status 03 (**Confirmed**), the system tries to book an appointment for the time slot that is defined by the appointment start date time and the appointment end date time. If this succeeds, a booked appointment with status 03 is created. Otherwise, the booking fails.